WB-700: Phase 1 Technical Design Document

Techfest 2026-27 · The Wrecking Ball Challenge · Phase 1 (Registration & CAD Abstract)

Autonomous, battery-powered, pendulum-driven spherical robot · CAD and system concept

1. Project title and team information

Field	Entry
Project title	**WB-700: autonomous pendulum-driven spherical robot** (working name, may be changed by the team)
Team ID / Team name	TM-47C20DF8A48 / StarDust
Institution	Madan Mohan Malaviya University of Technology, Gorakhpur
Team lead	Abhishek Kumar Singh | abhishekkrsingh2345@gmail.com | 8429339626
Members (max. 6)	Abhishek Kumar Singh (lead) · Gaurav Singh Rajpoot · Abhinav Rai · Naman Singh · Animesh · Ansh Pandey
Document	Phase 1 Technical Design Document, revision P1.0, 2026-10-24
Status	Concept design for the Round 1 design audit. **No hardware has been built or tested.**

All numbers come from the parametric CAD model (`cad/`) and its analysis scripts unless marked as an assumption, datasheet value or estimate. Placeholder hardware is labelled **PH**.

2. Executive summary

WB-700 is a **700 mm (2.297 ft) polycarbonate sphere** that moves by shifting its own centre of gravity. A gravity-held internal frame, the **stator**, hangs on a stationary hollow axle that runs between the two poles of the shell. Two 24 V gearmotors on the stator drive ring gears bolted to the shell's pole flanges. Their torque swings the stator forward, the CG moves ahead of the contact point and the ball rolls. A **steering bob**, made of the 6S Li-ion battery and laser-cut steel ballast arcs, hangs 45 mm below the axle on a longitudinal pivot. Tilting it ±30° leans the ball ±22.9° (static) to steer. A Raspberry Pi 5 runs all vision and navigation onboard. An RP2350-class MCU closes the 500 Hz motor, pendulum-damping and camera-stabilisation loops. An independent hardware safety chain implements the wireless kill switch.

Key results from the CAD model:

- **Dimensional compliance:** all calipers 696.3-700.7 mm (2.284-2.299 ft), strictly inside 2.0-2.5 ft, with 61.3 mm / 86.7 mm margins. No external sensors or mounts.
- **Mass and CG:** 17.00 kg in total (3.03 kg of it placeholders); **CG at (+1.4, -0.4, -87.2) mm** from the shell centre. The pendulum first moment is 1.48 kg m.
- **Packaging checks:** no clashes between 68 parts in 6 pitch/tilt states; at least 20.0 mm from the frame/bob to the shell at any pitch angle and over ±30° bob tilt. The camera view is 0.0 % obstructed in 9 bob/servo poses (CAD sightlines only). The battery can be removed without disturbing the ballast.
- **Preliminary ramp capability:** 10.7° steady climb including rolling resistance, with the baseline 4 ballast plates (7.1-13.6° with 2-6 plates). Conditional, because the ramp slopes are not published. For steeper ramps the energy for the 0.8 ft rise corresponds to a run-up of 1.31 m/s (20°) to 1.53 m/s (30°); this is an energy estimate only, not a demonstration of traction or crest control.

The design is original. The reference projects reviewed (SphBot, Rollie, Integral-Robotics-Lab Spherical-Robot) informed the choice of principle only; no CAD, code or text was reused.

3. Problem statement and competition objectives

The official problem statement (5 pages; checked against techfest.org on 2026-10-20 with no date conflict) asks for "*an autonomous, battery-powered spherical robot that navigates a 75-meter obstacle-laden arena using vision-based navigation, center-of-gravity stabilization, and real-time obstacle evasion.*"

Item	Requirement (official)	Design consequence
Geometry	Spherical, diameter strictly between 2.0 and 2.5 ft (609.6-762.0 mm)	700 mm nominal: margin on both sides

Item	Requirement (official)	Design consequence
Housing	All computing, power, actuators and electronics completely inside	Everything inside; the polar hubs are recessed
Peripherals	Sensors, mounts and lenses ≤ 0.5 ft (152.4 mm) beyond the shell	None used; camera inside
Power	Strictly onboard battery; no tether or remote power	6S1P Li-ion, about 97 Wh
Autonomy	100 % autonomous; no remote control, intervention or offboard processing	Pi 5 + MCU onboard; RF link is e-stop only
Sensors	Onboard camera primary; supplementary internal IMU, encoders, distance sensors for stabilisation	Camera primary; IMU + encoders internal
Safety	Operational wireless kill switch that immediately stops all motor actuation	Heartbeat e-stop, three independent cut channels
Track	S1 gate runway 0-15 m; S2 ramps 0.2/0.4/0.8 ft (15-35 m); S3 low-density obstacles (35-55 m); S4 swinging pendulum (55-75 m); red cones right, yellow left	Drive torque, traction, robustness, timing
Scoring	Run time + 5 s per cone contact; best of 3 attempts; DNF ranked by distance	Reliability first, then speed
Phase 1	CAD assembly with internal electronic layout, CoG placement, system schematic; judged on CAD, schematics, dimensional compliance	This document and the abstract

The official site also publishes a **Round 1 submission template** (Google Doc). The companion *Phase1_CAD_Abstract* follows it heading by heading.

4. Design requirements and compliance matrix

The full traceability matrix (40 rows: requirement, mandatory or design-dependent, design response, evidence, verification method, status) is in **Requirements_Compliance.csv**. Summary:

ID	Requirement (short form; exact wording in CSV)	Type	Status
R-01	Spherical, diameter strictly 2.0-2.5 ft	Mandatory	Satisfied
R-02	All compute, power, actuators, electronics inside	Mandatory	Satisfied
R-03	External sensors/mounts <= 0.5 ft beyond shell	Mandatory	Satisfied
R-04	Strictly onboard battery power	Mandatory	Satisfied
R-05	Phase 1 CAD + system document	Mandatory	Satisfied
R-06	Round 1 screening content	Mandatory	Partially satisfied
R-07	Phase 2 simulation report (future)	Mandatory	Partially satisfied
R-08	Round 2 audit content (future)	Mandatory	Partially satisfied
R-10	Inspection: size, weight, isolation, kill switch	Mandatory	Partially satisfied
R-11	Red cones right, yellow cones left	Design-dependent	Partially satisfied
R-12	S1 gate runway, vision calibration	Design-dependent	Partially satisfied
R-13	S2 ramps 0.2 / 0.4 / 0.8 ft	Design-dependent	Partially satisfied
R-14	S3 low-density obstacles, keep alignment	Design-dependent	Partially satisfied
R-15	S4 swinging pendulum	Design-dependent	Partially satisfied
R-19	Phase 1 deadline 25 Oct 2026	Mandatory	Partially satisfied
R-20	Register for Team ID	Mandatory	Unresolved
R-22	Original work; AI-tool rule	Mandatory	Unresolved
R-24	100 % autonomy, no offboard processing	Mandatory	Partially satisfied
R-25	Camera primary; internal supplementary sensors	Mandatory	Satisfied
R-26	+5 s per cone contact	Design-dependent	Partially satisfied
R-27	Wireless kill switch stops all actuation	Mandatory	Partially satisfied
T-01	Team identification	Mandatory	Unresolved
T-02	Isometric + cross-section renders	Mandatory	Satisfied
T-03	Shell material and exact OD	Mandatory	Satisfied

ID	Requirement (short form; exact wording in CSV)	Type	Status
T-04	0.5 ft rule confirmation	Mandatory	Satisfied
T-05	CoG coordinates + drive explanation	Mandatory	Satisfied
T-06	Motor torque calculation + selection	Mandatory	Partially satisfied
T-07	Electrical block diagram	Mandatory	Satisfied
T-08	Battery, BMS and LVC	Mandatory	Satisfied
T-09	Motor specs + driver boards	Mandatory	Partially satisfied
T-10	Kill switch + external manual disconnect	Mandatory	Satisfied
T-11	Software flowchart (state machine + vision stack)	Mandatory	Satisfied
T-12	Vision, estimation, Sector 4 logic	Mandatory	Partially satisfied

Derived design requirements (team targets, not rules):

ID	Derived requirement	Target	Status
DR1	Diameter margin to both limits, including forming tolerance	≥ 25 mm each side	86.7 / 61.3 mm: met
DR2	Moving parts to shell clearance (frame and bob)	≥ 8 mm	20.0 mm: met
DR3	Camera view free of internal obstructions	< 0.5 % blocked	0.0 % in 9 poses: met (CAD only)
DR4	CG below the shell centre with low lateral offset	z ≤ -60 mm, abs(y) ≤ 2 mm	z = -87.2, y = -0.4 mm: met
DR5	Quasi-static incline capability incl. rolling resistance (pendulum ≤ 60°)	≥ 12° (baseline)	10.7°: not met with 4 plates; 6 plates give 13.6°; ramp angle TBC
DR6	Battery below 100 Wh for transport	< 100 Wh	≈ 97 Wh: met
DR7	E-stop removes power from every actuator; fail-safe on signal loss	3 channels, latched	design: met; latency to be tested in Phase 2
DR8	Battery removable without disturbing CG hardware	path clear	clear (CAD envelope check)

5. Proposed robot concept

Parameter	Value	Basis
Shell	Clear PC, OD 700 mm, wall 3.0 mm (nominal)	design parameter
Total mass / CG	17.00 kg / (+1.4, -0.4, -87.2) mm	CAD mass table
Drive	Two-axis pendulum; 2 × 24 V gearmotors, 18T/72T m1.5 gears (4:1) to both shell poles	concept
Steering	Bob (5.34 kg) tilts ±30° about a pivot 45 mm below the axle; self-locking worm gearmotor	CAD
Compute	Raspberry Pi 5 (8 GB) + RP2350-class motor MCU	selected reference + PH
Primary sensor	Raspberry Pi Camera Module 3 Wide (102° × 67°), on a pitch-stabilising servo	datasheet
Supplementary sensors	9-DoF IMU, drive encoders, shell-index Hall, steering-angle sensor (all internal)	PH
Power	6S1P Li-ion 21700 (Molicel P45B), 21.6 V, 4.5 Ah, ≈97 Wh, BMS	datasheet
Safety	Heartbeat wireless e-stop; contactor + driver PWM gate + changeover brake relays; arming plug in left hub	concept

How it moves. On level ground at rest, the pendulum (10.15 kg, CG 146.2 mm below the axle) hangs vertically. A drive torque T between stator and shell rotates the stator forward by phi. The CG then moves forward by e·sin(phi), and the ball rolls to bring it back over the contact point. The largest torque the pendulum can react is m_p·e·g = 14.6 N m. **Steering:** tilting the bob by beta moves the CG sideways. The ball leans by gamma ≈ 22.9° at beta = 30° (CAD), and a leaning rolling sphere turns. **Vision:** the camera sits on the stator, so it does not spin with the shell. It looks forward through the clear shell 50 mm left of the seam plane, so the seam appears as a fixed, maskable stripe.

6. Justification for the selected internal drive mechanism

Three principles were compared (detail in `research/Mechanism_Research_and_Selection.md`):

Criterion (this track)	A. Two-axis pendulum	B. Internal drive unit (wheel cart)	C. Twin-hemisphere differential
Torque path	Gear into the shell, no slip	Wheel-to-shell friction; slips on ramps and impacts	Gear/bearing, but edge contact at the gap
Steering authority	Lean steering; no turn-in-place	Turns in place	Differential yaw
Barrier impacts	Pendulum cannot derail	Cart can climb or derail	Debris enters the equatorial gap
CG height achievable	Low (bob within 25 mm of the band)	Lowest	Moderate
Packaging and clearance proof	Pendulum sweeps a sphere about the centre: easy to prove	Cart, mast, sprung idlers	Large bearings at the equator
Vision	Stator camera through clear PC; seam masked	Views through the band the wheels scuff	Open gap, exposed to debris
Manufacturing	Plates, tube, stock bearings and gears	Moderate	Two precision large bearings
Score (/55, judgement)	**44**	36	38

Physics behind the choice. For any barycentre-offset drive the quasi-static incline limit is `sin(a_max) = (m_p e / (M R))·sin(phi_max)`. The required shell torque is `T = M g R (sin a + c_rr cos a)`. The pendulum must react T by gravity, so motor torque beyond `m_p e g` is useless. Climbing capability is therefore a *CG problem*, which the competition emphasises, and the pendulum architecture makes it explicit and tunable with ballast. Option B's theoretical advantage, a lower cart, depends on inner-shell friction μ ≥ tan(phi_cart) (≈0.84 at 40°). That is unreliable on clean or dusty polycarbonate and degrades exactly where the camera looks. On a linear 75 m course of sequential sectors (shape to be confirmed: Q4), turning in place is not needed; gate-to-gate corrections are, and lean steering provides ±22.9° of lean.

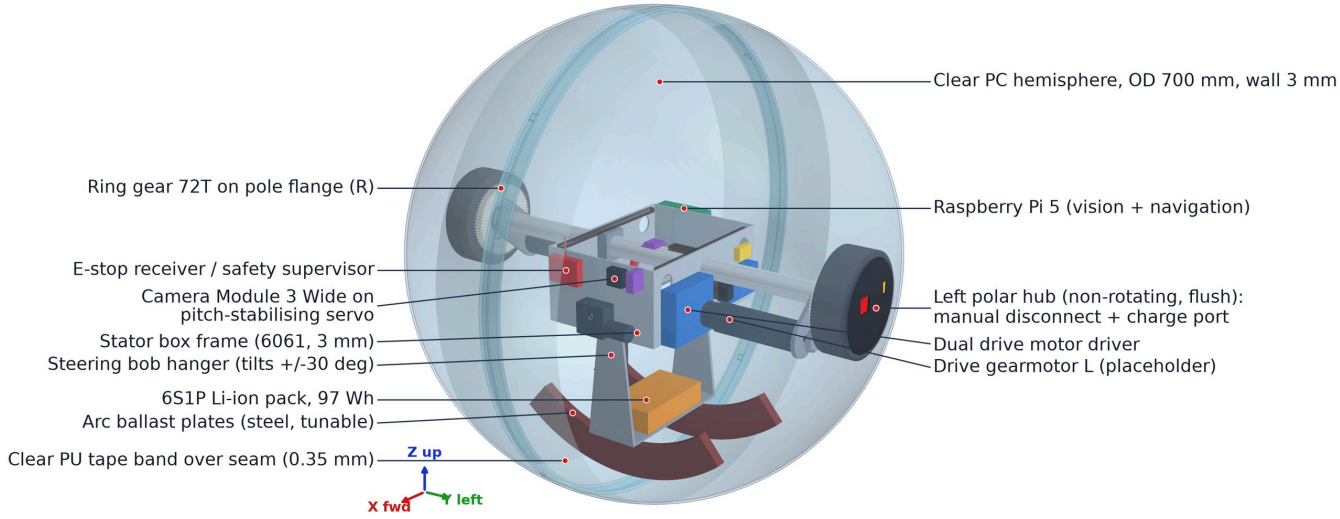
Reference projects (investigated, not copied): SphBot, a pendulum-driven sphere with split shell and gear drive (MIT, © 2019 A. Ahmadi; Kolbari et al., IcRoM 2018). Rollie, a 120 mm stator-and-shell ball with Camera Module 3 Wide and OpenSCAD sources (MIT, © 2025 D. M. Formoso). Integral-Robotics-Lab/Spherical-Robot, a pendulum ball with an ESP32-CAM (no licence file found, so nothing was reused). The commercial Rotundus GroundBot (pendulum drive, 60 cm, 25 kg) shows the principle at this scale.

7. Overall CAD design and dimensions

Workflow. The CAD is a parametric **CadQuery 2.8** (OpenCASCADE B-rep, the kernel FreeCAD also uses) Python model. `cad/wb700_params.py` holds every dimension and mass assumption. `cad/wb700_model.py` builds 68 named, coloured solids in five rigid bodies: shell, stator, bob, and envelopes. `cad/build_all.py` regenerates the analysis, the STEP assembly (opens in FreeCAD, Fusion, SolidWorks and others), per-part STL, a coloured GLB, and every figure. OpenSCAD and FreeCAD were not installed in the build environment, so a single executable source of truth was preferred over two parallel models.

WB-700 - isometric view (shell translucent)

Pendulum-driven ball | 700 mm shell | stationary-axle stator | twin geared drive | tilting ballast bob

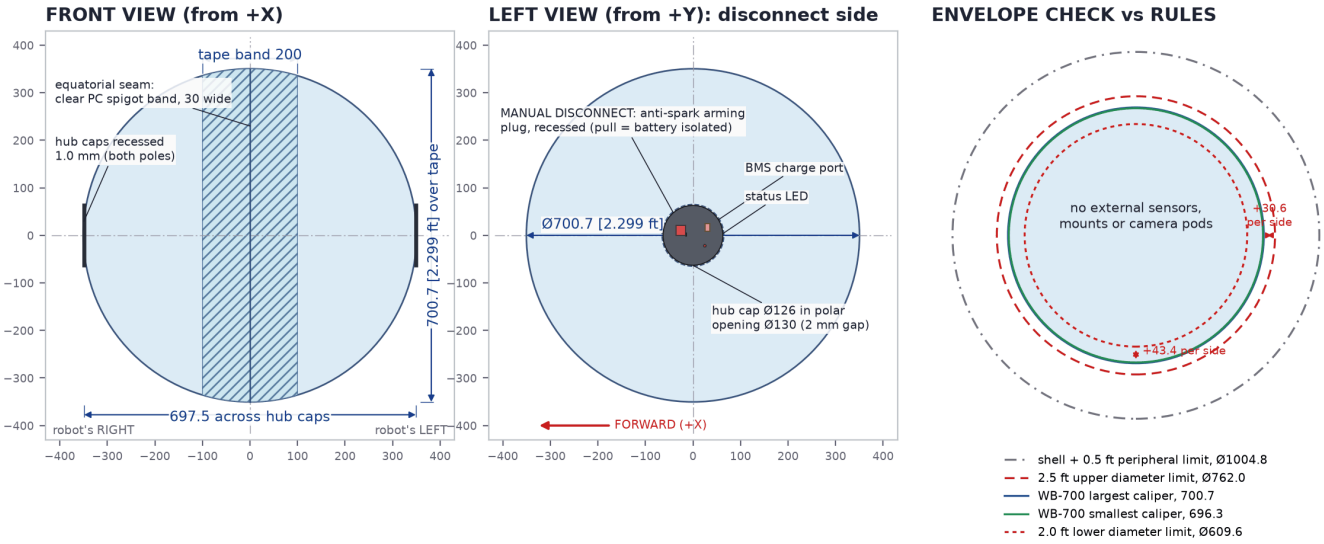


WB-700 Phase 1 concept CAD - generated by cad/build_all.py from wb700_params.py | grey/blue boxes = PLACEHOLDER envelopes for unselected purchased parts | units: mm

Figure 1 - Isometric render of the complete assembly (shell translucent).

WB-700 - external dimensions and envelope compliance

Exterior geometry from the CAD solids: PC shell, clear PU tape band, flush polar hubs and hub devices. Dimensions in mm [ft]. Outline radii measured on the CAD meshes.



Dimensional compliance (computed from the CAD meshes by wb700_analysis.dimensional_check)

Diameter rule (strict): 609.6 < D < 762.0 mm	D = 696.3 ... 700.7 mm (2.284 ... 2.299 ft), all 2562 sampled directions	PASS
Same rule with assumed +/-7 mm thermoforming tolerance	689.3 ... 707.7 mm	PASS
Peripheral limit: sensors/mounts/lenses <= 152.4 mm beyond shell	0 external items; largest non-skin feature -1.0 mm (ST-02, recessed)	PASS
Protective tape band counted in the diameter	+0.35 mm per side (included above)	INCLUDED
Margins to the rule limits	61.3 mm below 2.5 ft; 86.7 mm above 2.0 ft	INFO

Generated by cad/wb700_figures.py from the CadQuery solids in wb700_model.py (parameters: wb700_params.py). Units mm unless noted. Phase 1 concept - not for manufacture.

Figure 2 - External dimensions and envelope check (computed from the CAD meshes).

Key dimension	Value
Shell outer diameter / wall	700 mm / 3.0 mm PC
Diameter over tape band (max caliper)	700.7 mm (2.299 ft)
Minimum caliper (across recessed hubs)	696.3 mm (2.284 ft)

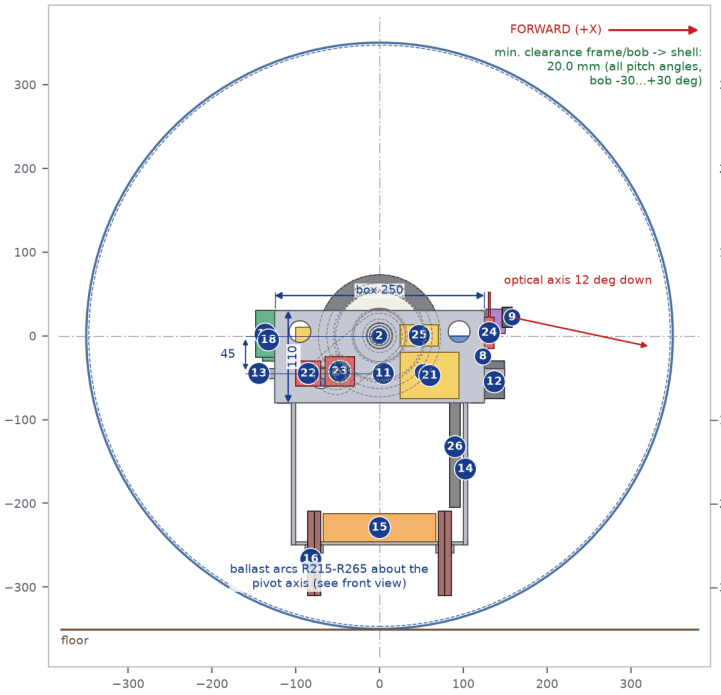
Key dimension	Value
Polar opening / hub cap	Ø130 mm / Ø126 mm, cap recessed 1.0 mm
Equatorial band	30 mm wide, 5 mm thick, clear PC
Axle	Ø30 × 3 mm 6061-T6 tube, 624 mm long, hollow harness duct
Pole bearings	61908-2RS (40 × 62 × 12 mm), 2×
Gear stage	72T / 18T, m1.5, centre distance 67.5 mm
Stator box	250 × 180 × 110 mm (6061 plates, 3 mm)
Steering pivot	45 mm below axle; ±30° range, rubber end-stops at ≈32°
Ballast arcs	R215-R265 mm about the pivot, ±40°, 8 mm steel, 4 plates (2-6)
Camera	lens at (158.4, 50, 22) mm, 12° look-down

8. Internal mechanical architecture

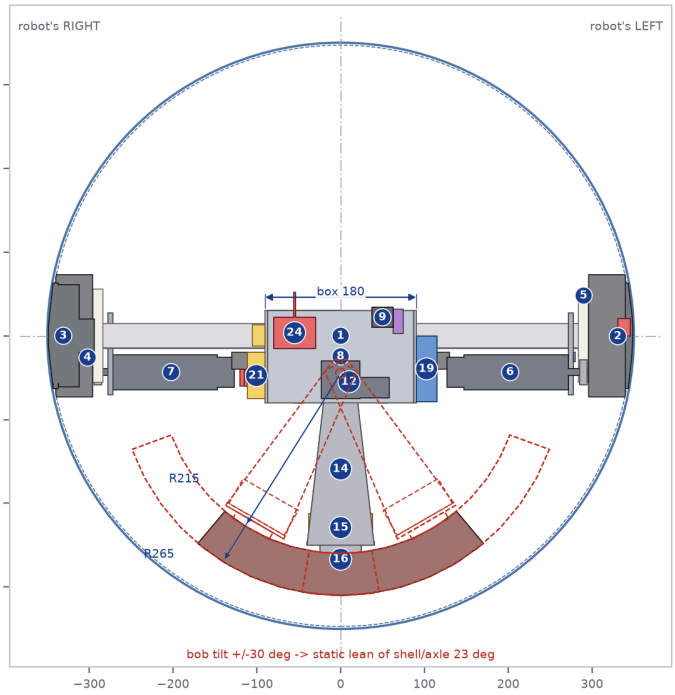
WB-700 - internal layout drawing (orthographic projections of the CAD solids)

Stator (gravity-held frame) and steering bob inside the 700 mm shell. PH = placeholder envelope for hardware not yet selected.

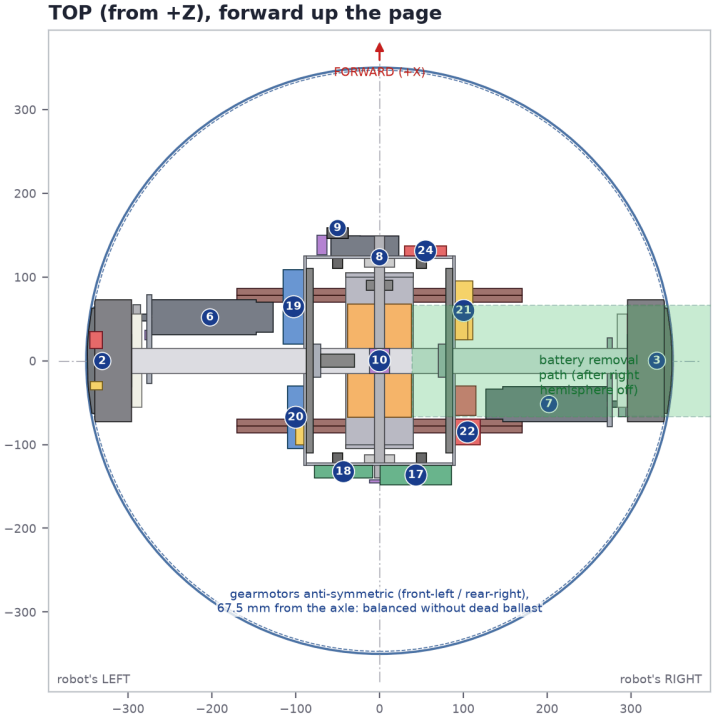
SIDE from the right (-Y); right hub + right drive dashed



FRONT (from +X); bob also at beta = +/-30 deg (dashed)



TOP (from +Z), forward up the page



Item list (mass: CAD volume x density, or datasheet/estimate for PH)

1	Stationary hollow axle 30x3 (6061-T6)	0.429 kg
2	Left polar hub: disconnect plug, charge port, LED	0.220 kg
3	Right polar hub + M12 preload bolt	0.224 kg
4	Pole bearings 61908-2RS (2x)	0.220 kg
5	Ring gears 72T m1.5 on pole flanges (2x)	0.188 kg
6	Drive gearmotor L, front-down (PH)	0.620 kg
7	Drive gearmotor R, rear-down (PH)	0.620 kg
8	Stator box frame: 4 plates + clamps	0.800 kg
9	Camera Module 3 Wide + pitch servo	0.004 kg
10	IMU 9-DoF on axle saddle (PH)	0.006 kg
11	Steering pivot shaft, 45 mm below axle	0.257 kg
12	Steering worm gearmotor (PH)	0.160 kg
13	Steering angle sensor (PH)	0.004 kg
14	Bob hangers (2x)	0.319 kg
15	Battery 6S1P 21700 + BMS (PH)	0.510 kg
16	Arc ballast plates, 8 mm steel (4x)	4.209 kg
17	Raspberry Pi 5 + cooler (PH)	0.070 kg
18	Motor-control MCU carrier (PH)	0.040 kg
19	Dual drive motor driver (PH)	0.090 kg
20	Steering motor driver (PH)	0.045 kg
21	Power distribution + current monitor (PH)	0.080 kg
22	Motor-power contactor, e-stop ch. A (PH)	0.040 kg
23	Dynamic-brake relays (PH)	0.060 kg
24	E-stop receiver + safety supervisor (PH)	0.030 kg
25	DC-DC 5.1 V / 5 A for Pi 5 (PH)	0.035 kg
26	Bob service loop (cable space)	0.080 kg

Generated by cad/wb700_figures.py from the CadQuery solids in wb700_model.py (parameters: wb700_params.py). Units mm unless noted. Phase 1 concept - not for manufacture.

Figure 3 - Internal layout drawing: orthographic projections of the CAD solids with the item list.

8.1 Shell and equatorial joint. Two thermoformed clear PC hemispheres meet on the plane y = 0. A clear PC spigot band is bonded inside the left rim and the right hemisphere slides over it. Four key tabs engage rim notches so the halves cannot rotate relative to each other. The joint is held closed by **axial preload through the axle**: one M12 bolt with disc springs in the right hub. There are no fasteners on the rolling surface. A sacrificial clear PU tape band (±100 mm) covers the seam and the contact band.

8.2 Polar interface. Each pole carries a PA12-CF pole flange (aluminium bearing insert) bonded and screwed to the shell around a Ø130 mm opening. The flange holds a 61908-2RS bearing and the 72T ring gear. The **stationary hubs** ride on the bearings' inner races and are keyed to the axle. Their caps sit flush in the openings with a 2 mm running gap and recessed 1.0 mm. The **left hub carries the manual disconnect, the charge port and the status LED.** Their cables run through the hollow axle, so no slip ring is needed.

8.3 Stator. Four 3 mm 6061 plates form a 250 × 180 × 110 mm box clamped to the axle by two split blocks. The box carries the electronics (Section 9), the steering pivot bearings, and the camera on its servo. The IMU sits on an axle saddle 22 mm above the pitch axis to limit tangential-acceleration error.

8.4 Twin geared drive. Each 42 mm-class planetary gearmotor (PH envelope 42 × 145 mm) bolts to an arm clamped on the axle. Its 18T pinion meshes with the ring gear of its own hemisphere, so the gear centre distance is referenced to the axle and bearing, not to the compliant shell. The anti-symmetric layout (left motor front-down, right motor rear-down) balances the stator fore-aft and sideways without dead ballast. It also puts drive torque into **both** hemispheres, so the equatorial joint carries little torque. When the right hemisphere is removed, its ring gear simply slides axially out of mesh.

8.5 Steering bob. Two 5 mm hangers hang from a Ø12 mm pivot shaft. They carry a 4 mm tray with the battery (removable along -Y) and two groups of **laser-cut steel ballast arcs centred on the pivot axis**. Because the arcs are concentric with the tilt axis, their clearance to the shell is set by their outer radius alone, and ballast mass is added in 8 mm (≈1.05 kg) steps. The self-locking worm gearmotor holds the bob angle without power, including after an e-stop.

8.6 Clearance and interference verification. Every shell-side internal part is a body of revolution about the axle, so stator pitch cannot change any clearance. Each stator and bob part was therefore checked once in the (radius, y) meridional plane against the shell, band, flanges, gears and bearings, for bob tilts of -30, -15, 0, +15 and +30°. Results: frame and bob to shell **≥ 20.0 mm** (worst: BL-01 vs SH-03 at beta 0). Precision polar gaps are ≥ 2.0 mm (hub cap to shell opening, by design). Exact B-rep intersections of every part pair, excluding designed contacts, were run in 6 states (pitch -60…+60°, tilt -30…+30°): **0 clashes**. Largest radius reached by the pendulum: 321.7 mm (inner shell radius 347 mm).

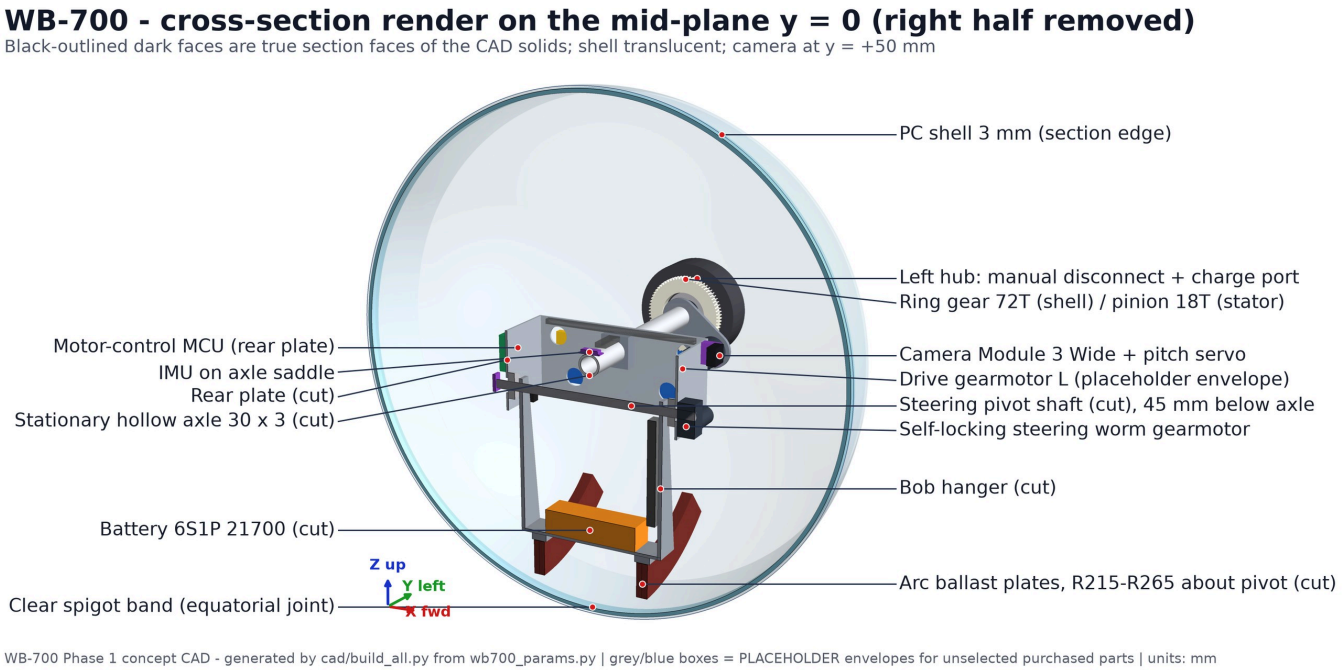


Figure 4 - Cross-section render on the mid-plane (true section faces outlined).

8.7 Impact management. Protection comes from several layers. The polycarbonate shell is tough and yields rather than shatters. The PU tape band protects the rolling and optical band. The heavy pendulum sits low and cannot derail. The steering bob has rubber end-stops. Electronics sit on standoffs on vibration-damping mounts. The battery uses

cylindrical cells in a rigid, padded tray instead of pouch cells. The drive torque is limited in software so a gear cannot be overloaded on impact. Drop and impact loads (0.8 ft ramp exit, barrier contact) are quantified in Phase 2.

8.8 Service and battery access. (1) Pull the arming plug. (2) Set the ball on its stand, left hub down. (3) Remove the single M12 preload bolt in the right hub. (4) Lift the right hemisphere: its ring gear slides out of mesh and its hub, bearing and flange come away with it. (5) Release the battery strap and slide the pack out along -Y; the CAD envelope check shows this path is clear. Charging is done in place through the left-hub port, so battery removal is a maintenance task, not a between-runs task.

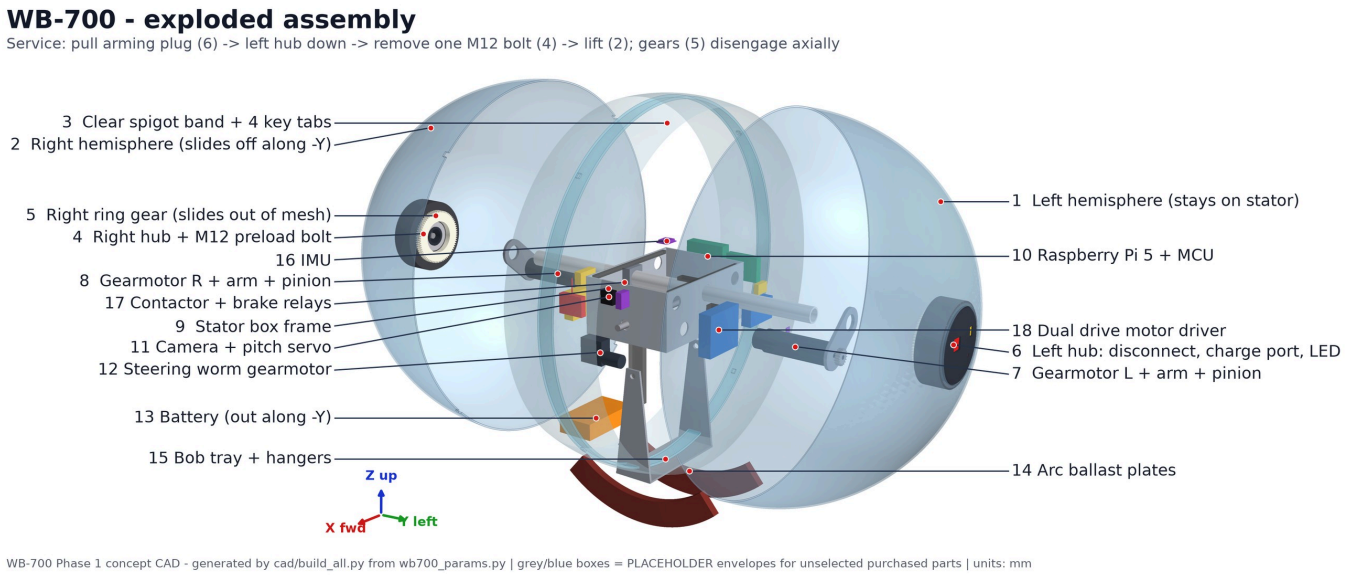


Figure 5 - Exploded assembly in service order.

9. Electronic component layout

ID	Component	CG (x, y, z) mm	Location	Mounting	Reason
SE-01	Camera Module 3 Wide (102 x 67 deg)	(152, 50, 22)	front plate, 50 mm left of seam plane, on servo	servo bracket	clear forward view; seam stripe off-centre
SE-02	Camera pitch-stabiliser servo (micro, metal gear)	(138, 69, 17)	front plate, upper left	screwed bracket	pitch-stabilises the camera
SE-03	IMU 9-DoF (BNO085-class) on axle saddle	(-0, 0, 17)	saddle on the axle, 22 mm above the pitch axis	clamp saddle	minimal tangential acceleration
SE-04	Steering angle sensor (AS5600-class magnetic)	(-144, 0, -45)	rear end of the pivot shaft	bracket off rear plate	absolute bob angle
SE-05	Shell-angle index (Hall sensor on L arm, magnet on ring gear)	(30, 279, -26)	left motor arm, facing ring-gear magnet	arm-mounted	once-per-revolution shell reference
CP-01	Raspberry Pi 5 (8 GB) + active cooler	(-136, -44, 2)	rear plate, outer face (right half)	M2.5 standoffs + damping	heat spread into the Al plate; away from the drivers
CP-02	Motor-control MCU carrier (Raspberry Pi Pico 2 class)	(-132, 43, -5)	rear plate, outer face (left half)	standoffs	short UART to the Pi; central harness
DV-01	Dual drive motor driver (Cytron MDD20A class)	(64, 102, -40)	left side plate, outer face, front	standoffs	short leads to both gearmotors
DV-02	Steering motor driver (Cytron MD10C class)	(-68, 100, -38)	left side plate, outer face, rear	standoffs	near the steering motor harness
PW-01	Power distribution board: fuse, INA228 monitor, bus bars	(60, -100, -48)	right side plate, outer face, front	standoffs	balances the left-side drivers
PW-02	DC-DC 24 V -> 5.1 V / 5 A (Pi 5 rail)	(48, -98, 0)	right side plate, outer face	standoffs	Pi rail kept separate from the motor bus
PW-03	DC-DC 24 V -> 6 V / 3 A (servo rail)	(-85, 95, 0)	left side plate, rear	standoffs	servo rail after the contactor
SA-01	Motor-power contactor (NO, 40 A) - e-stop channel A	(-85, -105, -45)	right side plate, rear	bracket	motor-bus isolation (channel A)

ID	Component	CG (x, y, z) mm	Location	Mounting	Reason
SA-02	Drive dynamic-brake relays + resistors (NC, fail-safe)	(-48, -102, -42)	right side plate, rear	bracket	dynamic brake (channel B)
SA-04	E-stop receiver + independent safety supervisor (ESP32-C3 class)	(131, -55, 4)	front plate, right; antenna up	standoffs	best RF view through the clear upper shell
PW-04	Manual disconnect: anti-spark arming plug (XT90-S class)	(25, 338, 10)	left polar hub face, recessed	panel mount	external manual disconnect
PW-05	Charge port (BMS charge input)	(-30, 338, 17)	left polar hub face	panel mount	charge without opening the ball
SA-03	Status LED (armed / e-stop / fault)	(-25, 338, -22)	left polar hub face	panel mount	armed / e-stop / fault indication
PW-10	Battery 6S1P 21700 Li-ion (Molicel P45B) + BMS	(-0, -0, -229)	steering-bob tray, lowest practical position	strap + rails	mass in the bob = steering authority

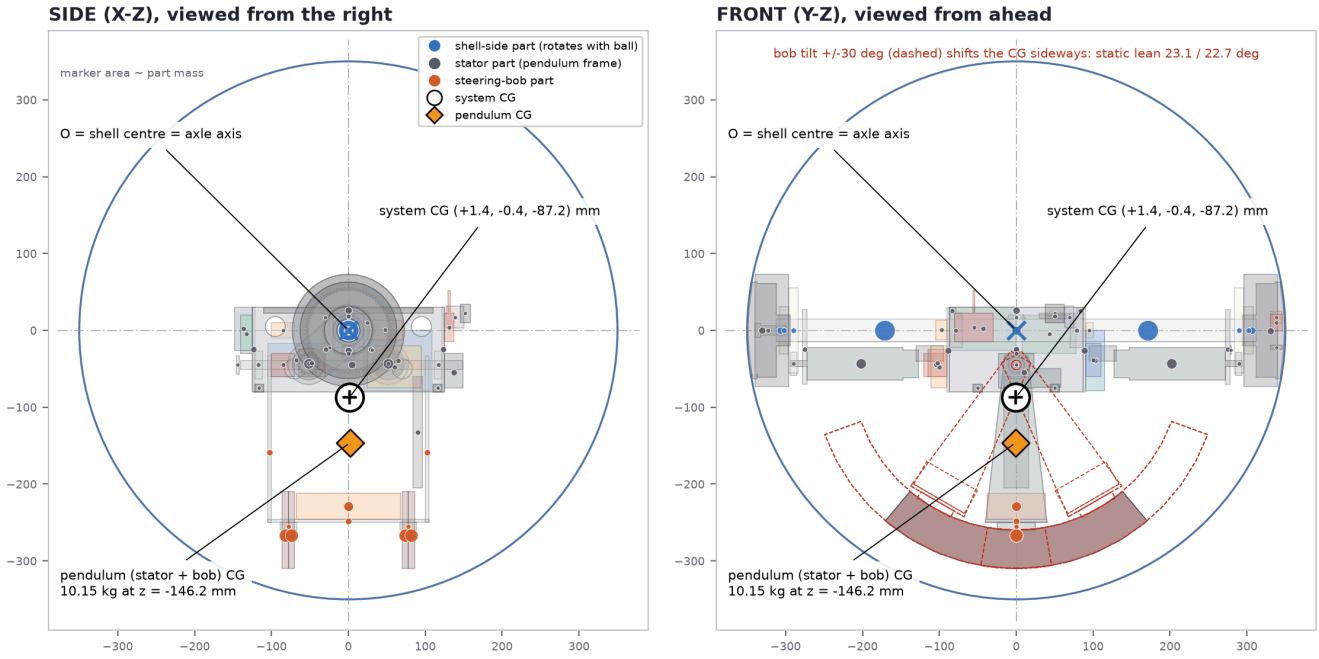
Cable routing. Harness trunks run along the inner faces of the side plates. The hub harness (arming plug, charge port, LED) runs inside the Ø24 mm bore of the stationary axle. Battery and steering-sensor leads reach the tilting bob through a service loop beside the front hanger, sized for ±30°. Motor leads run from the gearmotor end caps to the drivers on the side plates. Signal and motor power runs are separated by side: drivers on the left plate, power distribution on the right.

Thermal. The Pi 5 uses the official active cooler and is mounted on the aluminium rear plate, which acts as a heat spreader; the shell is sealed, so internal air temperature is monitored. A thermal estimate is part of Phase 2 (risk R11).

10. Center-of-gravity strategy

WB-700 - centre-of-gravity placement (static design estimate)

CG = sum(m_i r_i) / sum(m_i) over every CAD part (cad/out/mass_table.csv). Origin O = shell centre; X forward, Y left, Z up.



CG budget - static design estimate (dynamic behaviour is a Phase 2 result; hardware CG is measured in Phase 3)

Total mass (all CAD parts; baseline 4 ballast plates)	17.00 kg, of which 3.03 kg in 27 placeholder items
System CG relative to O (x, y, z)	(+1.4, -0.4, -87.2) mm: 87.2 mm below the shell centre
Shell body / stator / bob	6.86 / 4.81 / 5.34 kg
Pendulum first moment m_p e about the axle	1.483 kg m (climb index m_p e / M R = 0.249)
Quasi-static incline limit, pendulum at 60 / 90 deg	10.7 / 12.7 deg (ballast 2..6 plates: 7.1..13.6 deg at 60 deg)
Stator rest pitch from the fore-aft CG offset	+0.91 deg (trimmed by camera servo)
Static lean at bob tilt 30 deg	22.7 deg

Generated by cad/wb700_figures.py from the CadQuery solids in wb700_model.py (parameters: wb700_params.py). Units mm unless noted. Phase 1 concept - not for manufacture.

Figure 6 - CG placement from the CAD mass table.

Method. For every part i, mass m_i comes either from the CAD volume × material density (structural parts) or from a datasheet value or estimate (purchased parts, flagged PH). The position r_i is the centroid of the part's solid. Then `r_CG = Σ m_i r_i / Σ m_i`. The same calculation is repeated for the stator and bob alone, and for every pitch and tilt state. The table is reproducible from `cad/out/mass_table.csv`, and `Preliminary_BOM.csv` aggregates it.

Mass group	kg	%	x mm	y mm	z mm	CAD / est. kg
Shell + equatorial joint + tape	5.972	35.1	+0.0	-0.0	+0.0	5.97 / 0.00
Polar interface (flanges, bearings, hubs, hub devices, bolt)	1.222	7.2	+0.4	-2.3	+0.2	0.96 / 0.26
Twin drive (gearmotors, pinions, arms, ring gears)	1.663	9.8	-0.0	-0.0	-36.6	0.42 / 1.24
Stator structure (axle, plates, clamps, fasteners)	1.478	8.7	+0.0	+0.0	-7.6	1.23 / 0.25
Steering mechanism (pivot, bearings, worm motor, stops, hangers, tray, brackets)	1.114	6.5	+20.7	+1.4	-134.9	0.76 / 0.35
Ballast arc plates	4.209	24.8	+0.0	+0.0	-266.8	4.21 / 0.00
Battery 6S1P + BMS	0.510	3.0	+0.0	+0.0	-229.0	0.00 / 0.51
Compute (Pi 5, MCU)	0.110	0.6	-135.0	-12.0	-0.5	0.00 / 0.11
Motor drivers	0.135	0.8	+20.5	+101.7	-39.2	0.00 / 0.14
Power electronics (PDB, DC-DC)	0.130	0.8	+39.9	-77.1	-29.1	0.00 / 0.13

Mass group	kg	%	x mm	y mm	z mm	CAD / est. kg
Safety hardware (contactor, brake relays, supervisor)	0.130	0.8	-17.8	-92.3	-32.6	0.00 / 0.13
Camera, servo and sensors	0.030	0.2	+67.8	+57.5	+6.7	0.00 / 0.03
Wiring harness (allowance)	0.300	1.8	+24.0	+8.3	-27.0	0.00 / 0.30
Total	**17.003**	100.0	+1.4	-0.4	-87.2	13.56 / 3.44

Result: CG = (+1.4, -0.4, -87.2) mm, i.e. 87.2 mm below the centre. Pendulum first moment m_p·e = 1.48 kg m; climb index m_p e / (M R) = 0.249. The fore-aft offset tilts the stator 0.91° at rest, which the camera servo trims.

Influence on behaviour (static estimates; dynamics are validated in Phase 2):

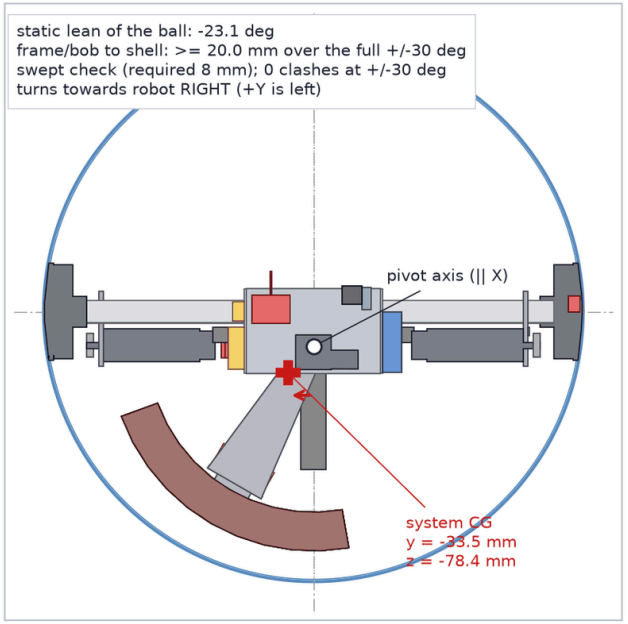
- *Ground-contact stability:* the contact point is always directly below the sphere centre. Any CG offset produces a restoring moment of M·g·87.2 mm × sin(tilt), so the ball rights itself when disturbed.
- *Ramp climbing:* capability scales with m_p·e/(M·R). Including rolling resistance, the baseline gives a steady climbing limit of 10.7° (pendulum ≤ 60°) and the ballast range gives 7.1-13.6° (Section 14). This is conditional: the ramp heights are published, their slopes are not.
- *Acceleration and braking:* the stator swings by phi ≈ asin(M_eff·a·R / (m_p·e·g)). An acceleration of 0.5 m/s² needs about 15° of swing, inside the camera-servo range. The MCU damps the swing.
- *Steering:* the bob holds 53 % of the pendulum mass and hangs 211 mm below its pivot, so ±30° of tilt gives about ±22.9° of static lean (CAD, Figure 7). At both ±30° extremes the frame and bob stay at least 20.0 mm from the shell with no clashes. Turning radius and reliable gate navigation are Phase 2 dynamic results.
- *Disturbance recovery:* the low CG gives a strong restoring moment. Pitch oscillation (pendulum mode) is damped by the motor loop using IMU pitch rate.
- *Barrier contact:* the heavy, low pendulum keeps the contact force high and resists the yaw torque from off-centre hits. Gyro heading-hold corrects residual yaw. No internal part can derail.

These are static estimates. They do **not** guarantee stability; that is shown in Phase 2 (dynamic model) and Phase 3 (measured CG by suspension test).

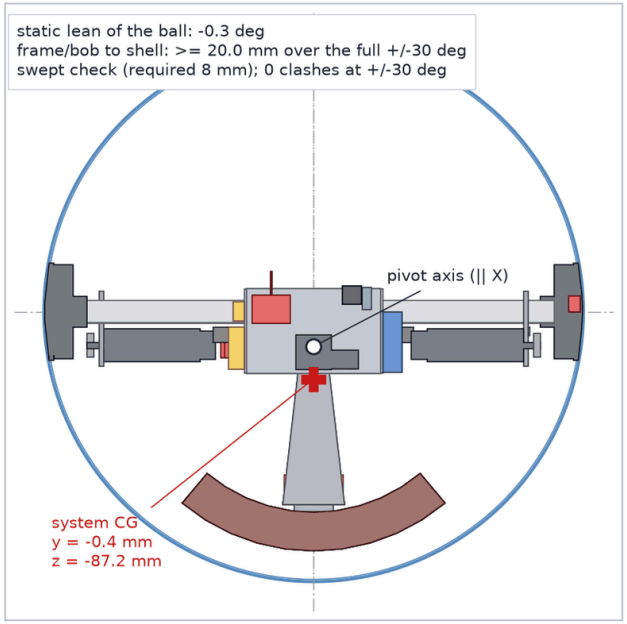
WB-700 - steering by bob tilt (front view, looking backwards at the robot)

A self-locking worm gearmotor tilts the bob about a pivot 45 mm below the axle. The CG moves sideways, the ball leans until the CG is over the contact point, and it turns towards the lean. Static CAD estimate.

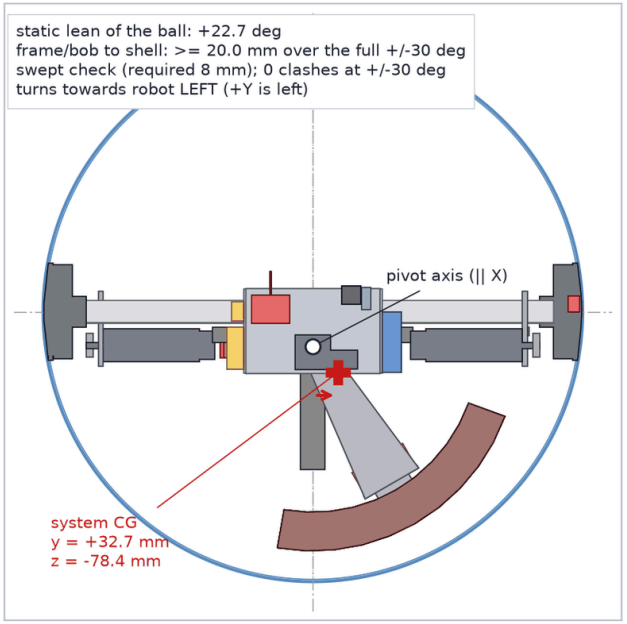
bob tilt beta = -30 deg



bob tilt beta = +0 deg



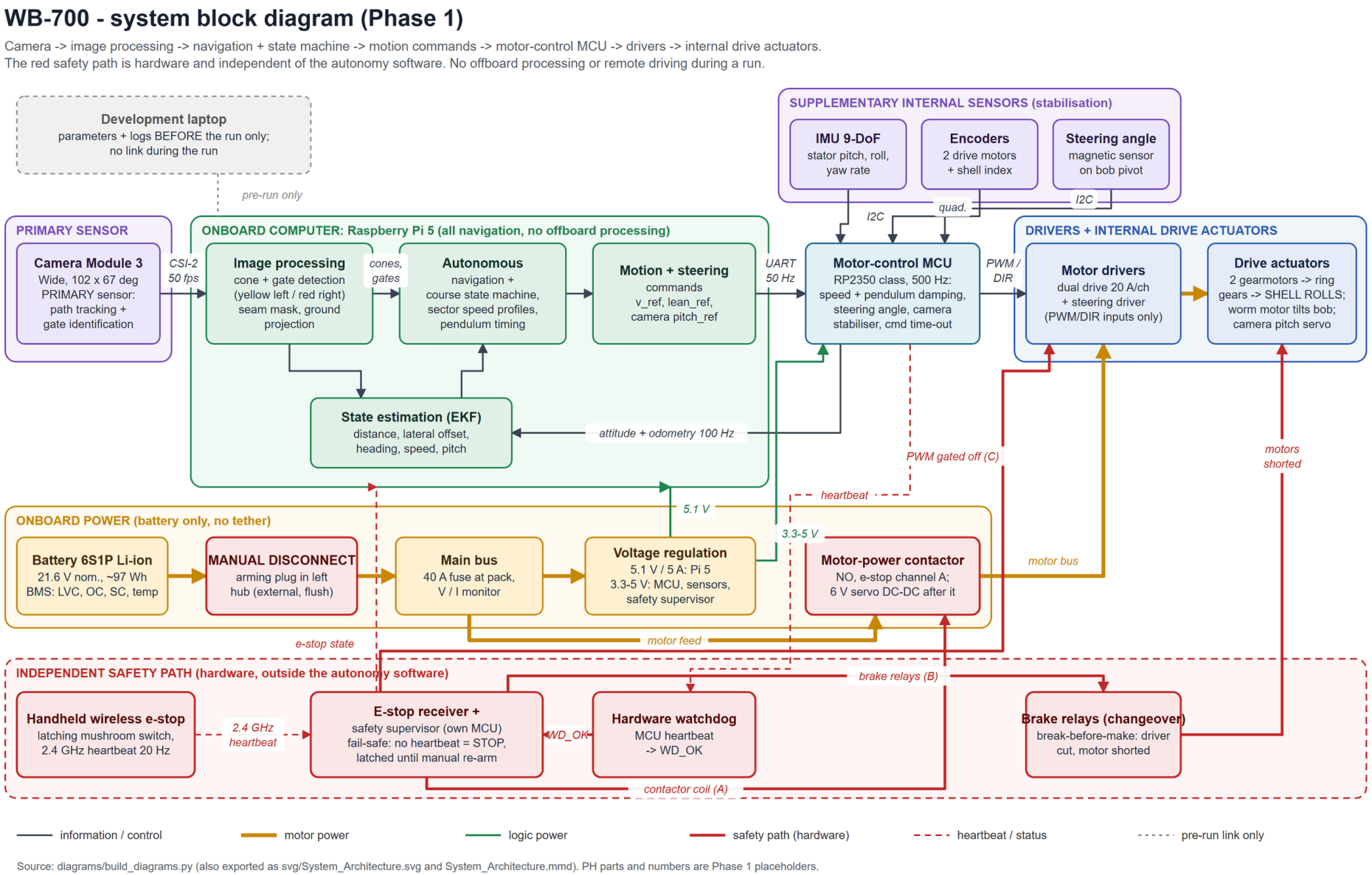
bob tilt beta = +30 deg



Lean is a static estimate (CG directly above the contact point). Turning radius versus speed and reliable gate navigation are Phase 2 dynamic results.
Generated by cad/wb700_figures.py from the CadQuery solids in wb700_model.py (parameters: wb700_params.py). Units mm unless noted. Phase 1 concept - not for manufacture.

Figure 7 - Steering by bob tilt at -30°, 0° and +30°: pivot axis, sideways CG shift, static lean and clearance to the shell.

11. System block diagram



Source: diagrams/build_diagrams.py (also exported as svg/System_Architecture.svg and System_Architecture.mmd). PH parts and numbers are Phase 1 placeholders.

Figure 8 - System block diagram: information flow, power, and the independent safety path (red).

How the system operates. (1) **Power-up:** the arming plug is inserted and the logic rail boots the Pi 5, MCU and safety supervisor. The MCU self-tests its sensors, encoders and drivers with motor power still off. (2) **Arm:** the operator sets the e-stop transmitter to RUN. The supervisor closes the contactor and energises the brake relays (motors connected to the drivers) only if the heartbeat is valid and WD_OK is high. (3) **Calibrate:** in Sector 1 the vision node locks colour thresholds and exposure on the chevron gates while rolling slowly. (4) **Run:** the Pi 5 turns images into a lane model. The EKF fuses it with IMU and encoder data, and the state machine selects the sector behaviour and sends v_ref and lean_ref at 50 Hz over UART. The MCU closes drive-speed, pendulum-damping, bob-angle and camera loops at 500 Hz. If commands stop for more than 200 ms the MCU makes a controlled stop. (5) **Stop:** at the finish, on any software fault (SAFE_STOP), or on the hardware e-stop, which acts on all three channels and latches.

The development laptop is used only before the run, to set parameters and download logs. It is disconnected during the run, so there is no remote driving and no offboard processing.

12. Preliminary power and control architecture

WB-700 - preliminary power distribution

Strictly onboard battery power. Every actuator (drive, steering, camera servo) is downstream of the e-stop contactor; the logic rail stays alive so the robot can report and log the stop. Values are Phase 1 design targets; fuse and wire sizes are confirmed in Phase 2.

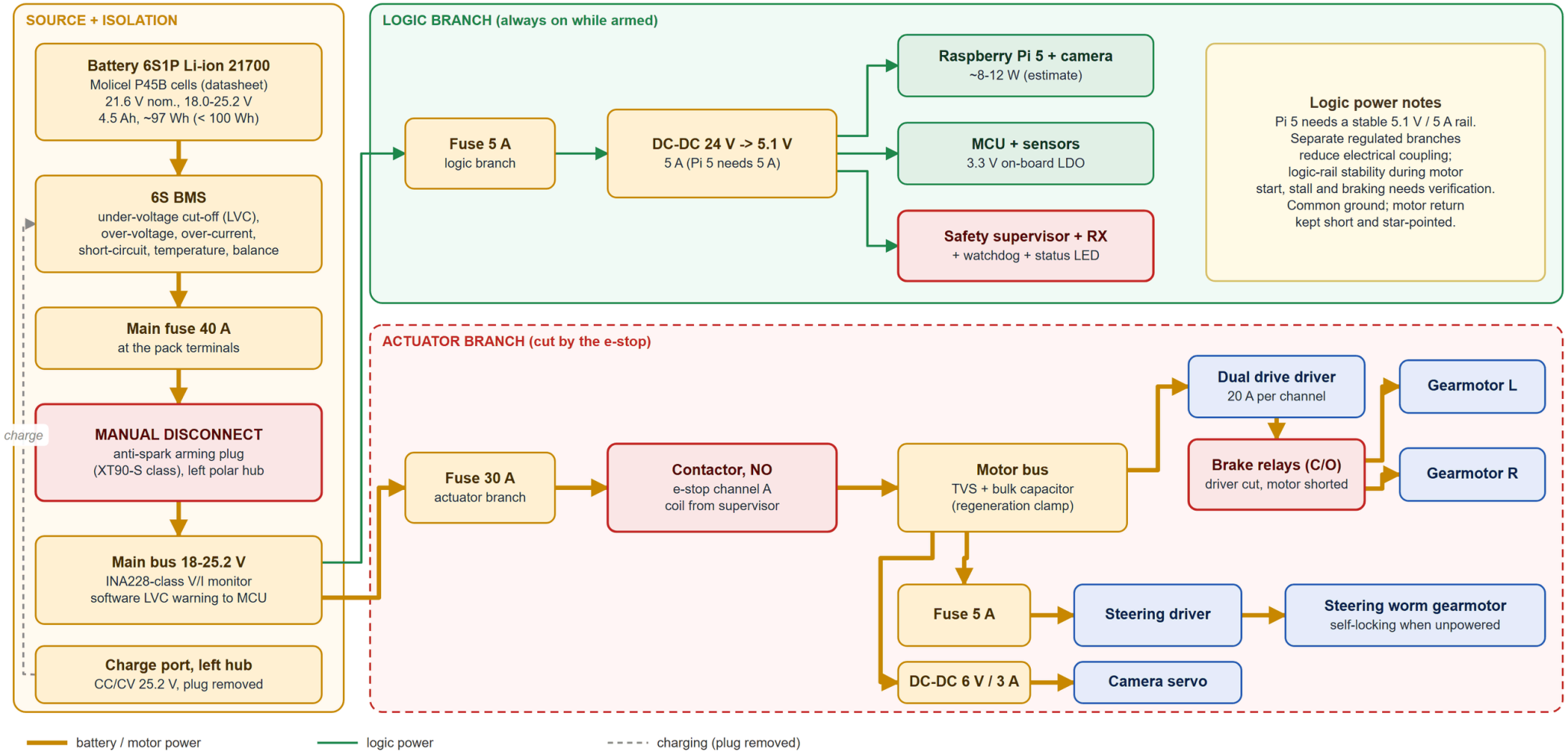


Figure 9 - Preliminary power distribution: every actuator is downstream of the e-stop contactor.

WB-700 - preliminary electronics architecture (boards + interfaces)

PH = placeholder part class (final selection in Phase 2). Buses: CSI-2 camera, UART/USB Pi-to-MCU, I2C/SPI sensors, PIO quadrature encoders, PWM/DIR drivers.

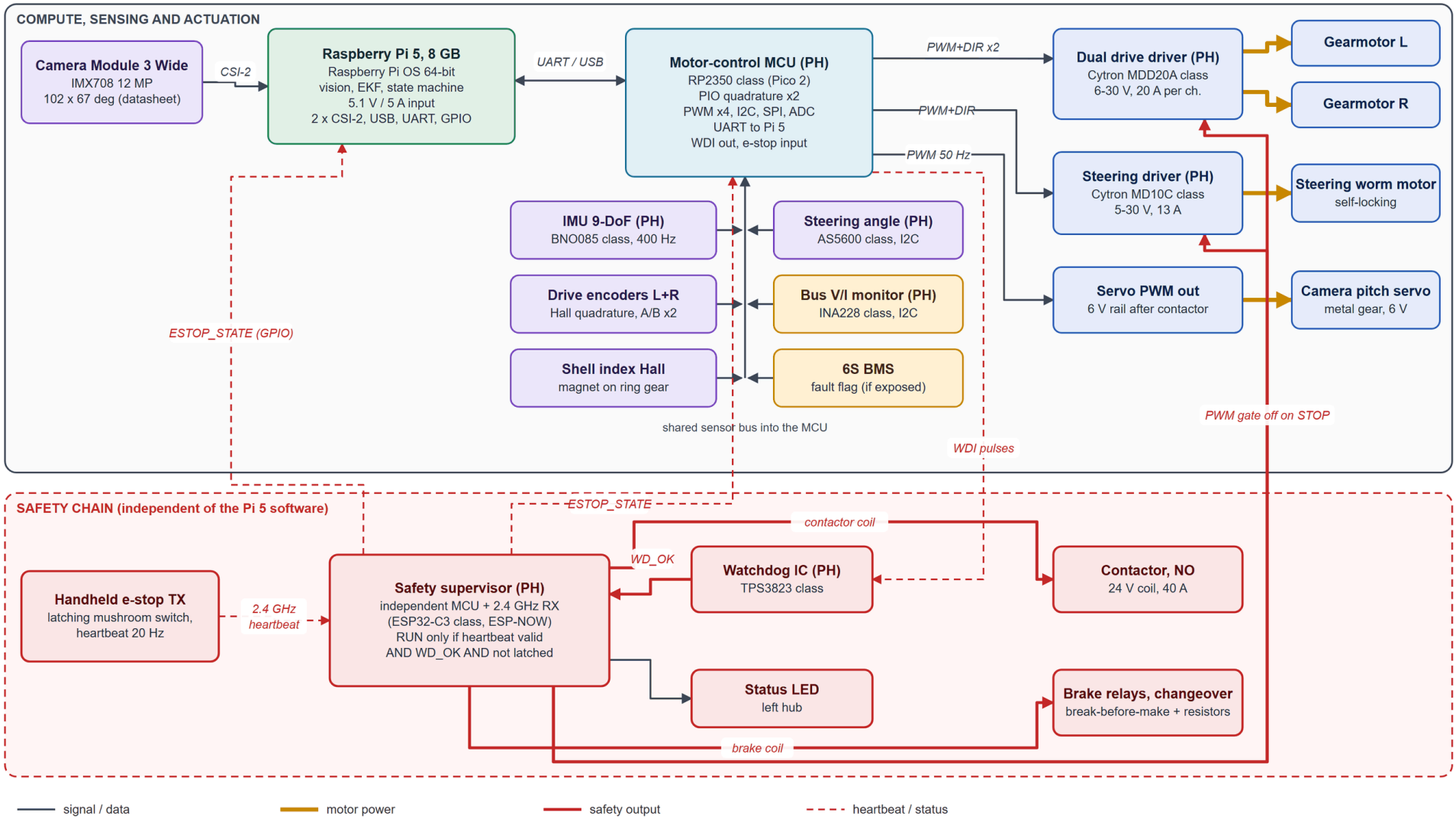


Figure 10 - Preliminary electronics architecture (boards and interfaces).

Component-interface table

Link	Physical layer / protocol	Rate	Level / note
Camera -> Pi 5	MIPI CSI-2, 2 lanes (15-to-22-pin FPC)	1280x720 @ 50 fps	camera-side 3.3 V
Pi 5 <-> MCU	UART 1 Mbaud (or USB-CDC)	cmd 50 Hz / telemetry 100 Hz	3.3 V logic
IMU -> MCU	I2C 400 kHz or SPI	400 Hz	3.3 V
Drive encoders -> MCU	quadrature A/B on PIO	up to ~50 kHz edges	level-shift if 5 V
Shell index Hall -> MCU	digital interrupt	1 pulse / shell rev.	3.3 V
Steering angle sensor -> MCU	I2C (AS5600 class)	200 Hz	3.3 V
Bus monitor -> MCU	I2C (INA228 class)	50 Hz	3.3 V
MCU -> drive driver	PWM 20 kHz + DIR, x2	500 Hz update	3.3 V logic
MCU -> steering driver	PWM + DIR	500 Hz update	3.3 V logic
MCU -> camera servo	servo PWM 1-2 ms	50 Hz	signal 3.3 V, power 6 V
MCU -> watchdog IC	WDI toggle	>= 10 Hz	3.3 V
Watchdog -> supervisor	WD_OK, open drain	level	3.3 V
E-stop TX -> supervisor	2.4 GHz (ESP-NOW class) heartbeat	20 Hz, 250 ms time-out	RF
Supervisor -> contactor	low-side MOSFET coil driver + flyback diode	level	24 V coil
Supervisor -> brake relays	coil driver (NC contacts short motors)	level	24 V coil
Supervisor -> drivers	PWM gate (AND + pull-downs)	level	3.3 V
Supervisor -> MCU, Pi 5	ESTOP_STATE GPIO	level	3.3 V
DC-DC -> Pi 5	5.1 V / 5 A to the 5 V input	continuous	Pi 5 USB current limit set in config

Control loops (targets): drive speed and pendulum-pitch damping 500 Hz (MCU); steering-bob angle 500 Hz (MCU); camera pitch stabiliser 100 Hz (MCU); EKF 100 Hz (Pi 5); vision 30-50 Hz (Pi 5); state machine and planners 50 Hz (Pi 5).

Energy (estimate, validated in Phase 2): a ≈97 Wh pack against roughly 10-15 W of electronics (Pi 5 + camera, estimate) plus about 40-80 W average drive power while moving (estimate). A 2-minute run uses only a few Wh, so three attempts and practice runs fit comfortably on one charge.

13. Vision-based navigation concept

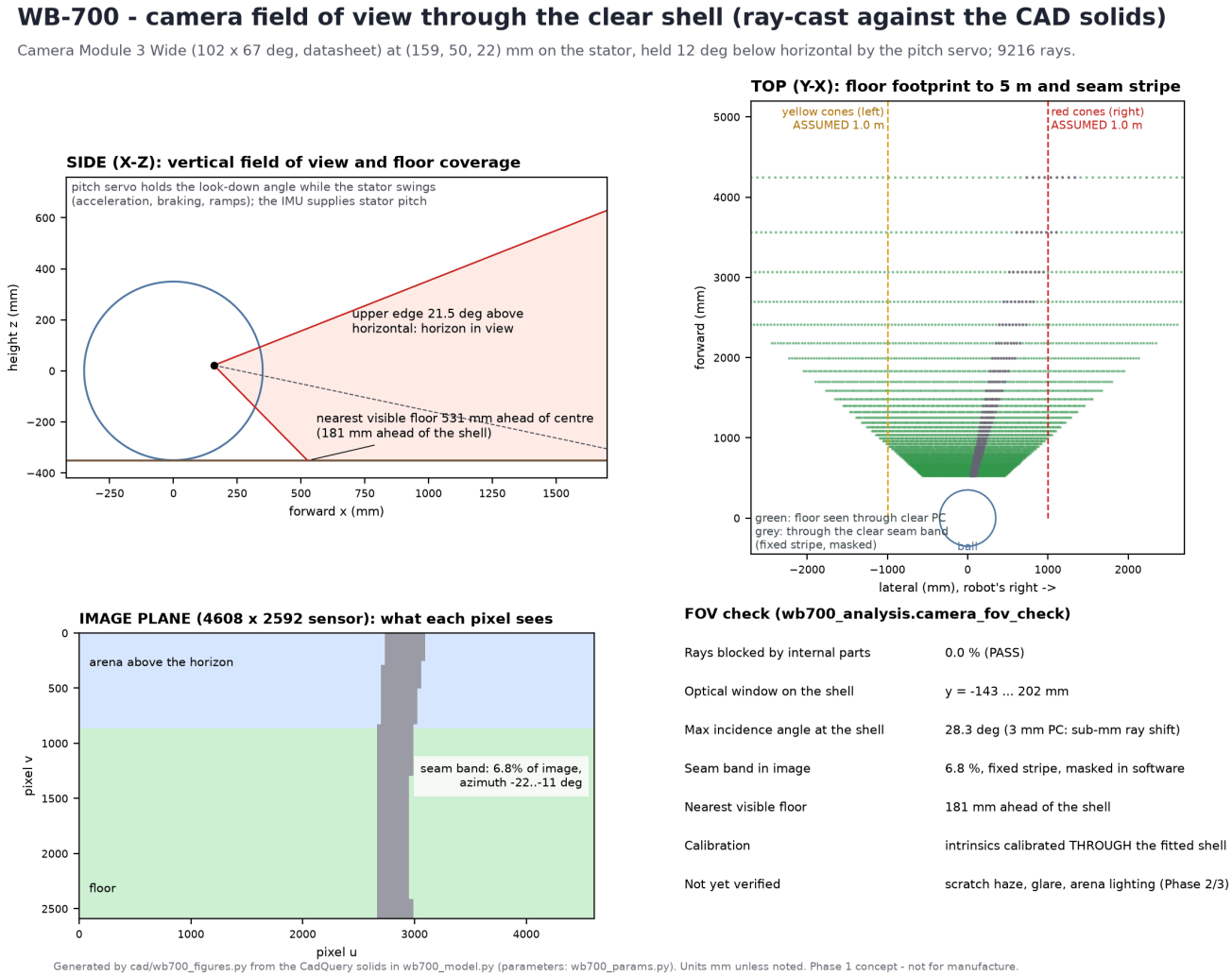


Figure 11 - Camera field of view ray-cast against the CAD solids.

- **Sensor:** Raspberry Pi Camera Module 3 Wide (IMX708, 102° × 67°, 2.75 mm, f/2.2, from the datasheet) at 1280 × 720 and 50 fps. Its wide vertical view keeps the near track visible (181 mm ahead of the shell) and the horizon in view.
- **Through-shell imaging:** the camera sits 159 mm ahead of the centre, so rays meet the 3 mm shell at ≤ 28.3° incidence and are displaced by less than a millimetre. Intrinsics are calibrated with the shell installed. The clear seam band covers 6.8 % of the image as a fixed stripe at -21.9° to -10.6°, which is masked. Interior surfaces near the camera are matte black to limit reflections.
- **Pipeline (Figure 12):** rectify → IMU/servo ground homography → adaptive HSV segmentation of red and yellow cones → blob filtering and ground projection → RANSAC boundary model (yellow = left, red = right) → lane centre, heading error and gate detection → EKF. An optional YOLO-nano-class detector (CPU, or an AI accelerator HAT if added) cross-checks cones and detects the pendulum.
- **Robustness:** auto-calibration on the Sector 1 chevron gates; a lane-loss timer (> 0.5 s triggers SAFE_STOP); gyro and odometry dead-reckoning bridge short occlusions in the wrecking zone.

WB-700 - software flowchart: vision / AI stack and course state machine

Concept level for Phase 1. Everything runs on the Raspberry Pi 5; the MCU closes the 500 Hz loops. Algorithms are developed and validated in Phase 2.

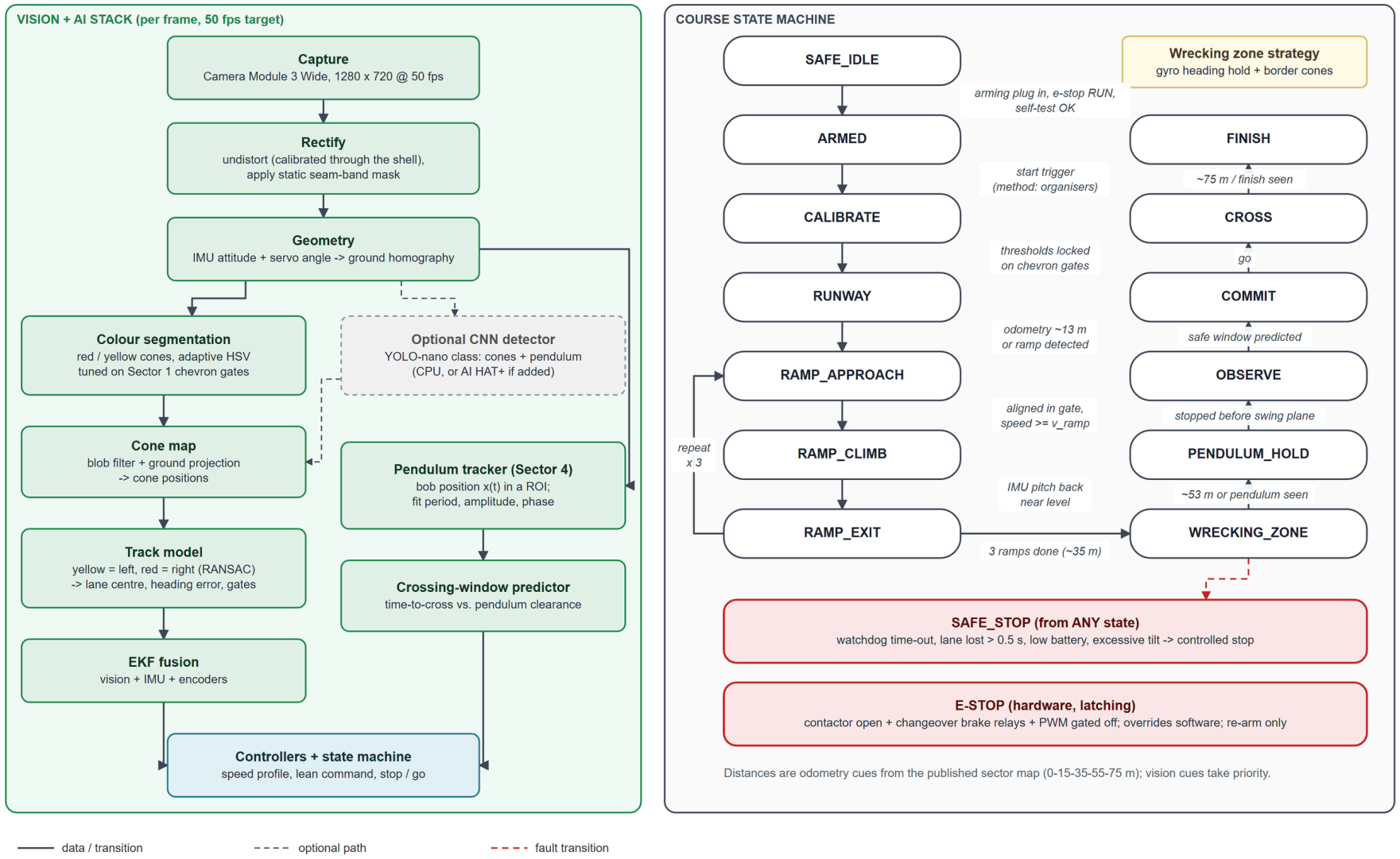


Figure 12 - Vision / AI stack and course state machine (concept).

14. Approach to ramps and low-density obstacles

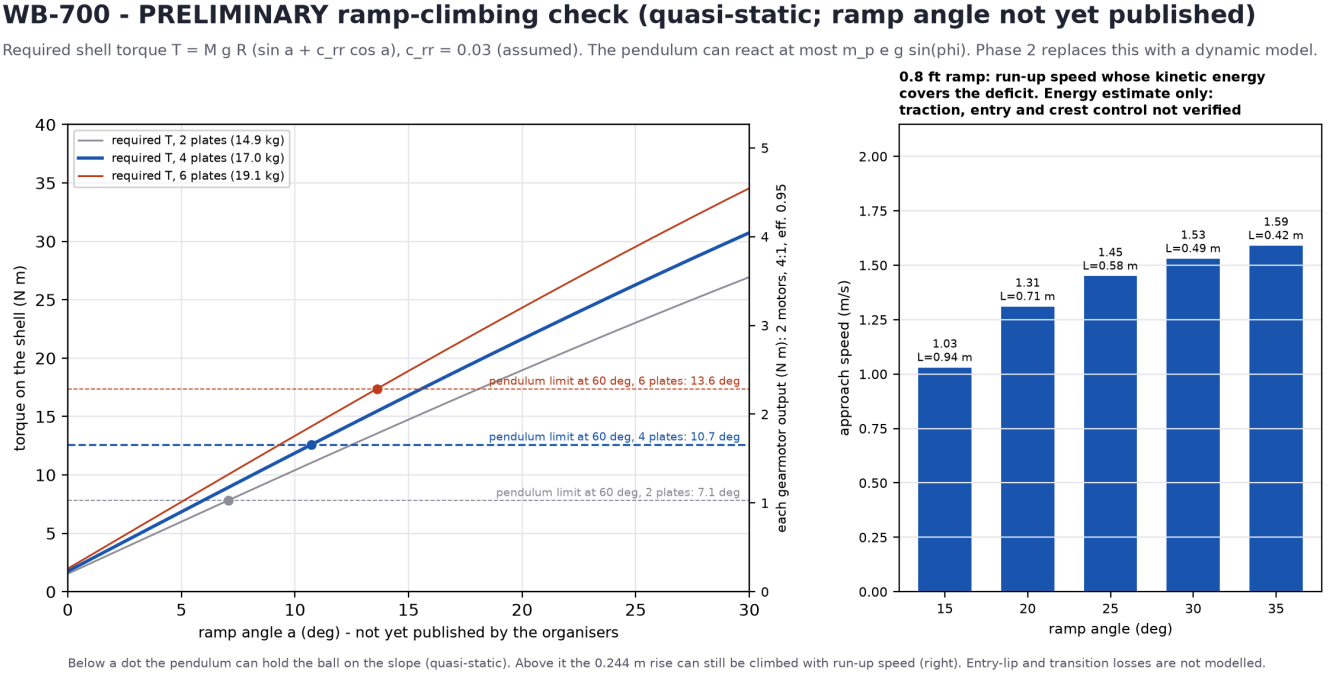


Figure 13 - Preliminary ramp check (quasi-static) and run-up speeds for the 0.8 ft rise.

Ramp angle (deg)	Shell torque (N m)	Per gearmotor (N m)	Pendulum angle (deg)	Quasi-static (pendulum <= 60 deg)
5	6.8	0.90	28.0	yes
8	9.9	1.30	42.7	yes
10	11.9	1.56	54.7	yes
12	13.9	1.82	72.3	run-up needed
14	15.8	2.08	-	run-up needed
16	17.8	2.34	-	run-up needed
20	21.6	2.85	-	run-up needed
25	26.3	3.46	-	run-up needed
30	30.7	4.04	-	run-up needed

Ramps. In RAMP_APPROACH the robot centres in the funnel gate, aligns its heading with the gyro and reaches v_ramp. In RAMP_CLIMB it switches to torque mode with the pendulum angle capped at 60°, so it cannot loop over. Ramps steeper than the 10.7° steady limit need run-up energy (table below). The table is an energy estimate only: it does not show that traction holds, that the entry transition is survivable or that the robot stays controllable at the crest.

Motor operating points (per motor, gearmotor output, 4:1 at 95 %): steady climb at 10.7°: 1.66 N m at 55 rpm (0.5 m/s, assumed); cruise at 2.0 m/s: 0.23 N m at 218 rpm; pendulum horizontal: 1.91 N m, short-term. Target motor at 24 V: rated (continuous) ≥ 2 N m, stall ≥ 4 N m, no-load ≥ 320 rpm. At the 18.0 V minimum pack voltage this leaves about 3.0 N m stall, 240 rpm unloaded and about 107 rpm at the climb torque.

Drive-motor selection evidence (selection pending). No motor model is chosen yet; the table states what the chosen part must show on its datasheet at the same operating point.

Item	Requirement	Selected motor
Model	42 mm-class 24 V brushed planetary gearmotor with Hall encoder	**pending (Phase 2)**
Supply for the check	18.0 V minimum pack voltage (6S at cut-off)	pending
Climb point	1.66 N m continuous at ≥ 55 rpm at 18 V	pending
Cruise point	0.23 N m at ≥ 218 rpm at 18 V	pending
Peak	1.91 N m short-term (pendulum horizontal)	pending

Item	Requirement	Selected motor
Current	continuous current at the climb point ≤ driver 20 A/ch continuous; stall ≤ 60 A peak	pending
Thermal	climb point sustainable for ≥ 30 s (longest ramp approach) within rated winding temperature	pending

In RAMP_EXIT it brakes with pendulum damping, because what lies beyond the top (down-slope or drop) is unknown (Q3). The design levers are ballast (2-6 plates), shell diameter (a parameter) and the gear ratio.

Ramp angle (deg)	Ramp length for a 0.244 m rise (m)	Approach speed (m/s)
15	0.94	1.03
20	0.71	1.31
25	0.58	1.45
30	0.49	1.53
35	0.42	1.59

Low-density obstacles (Sector 3). Foam, thermocol and cardboard are pushed or rolled over rather than avoided. The robot keeps 1-1.5 m/s for momentum and holds heading on the gyro while border cones remain visible. Nothing inside can derail on impact, and the PC shell and tape band take the abrasion. A quasi-static lip of about 11 mm can be mounted from standstill; larger debris is crossed dynamically (run-up), which Phase 2 quantifies.

15. Pendulum-detection and safe-crossing concept

1. **Detect:** at about 53 m (odometry), or when the vision node flags a periodic mover crossing the lane, enter PENDULUM_HOLD and stop at a hold line before the swing plane. A pendulum robot must be *actively* held, so the MCU uses encoder and IMU station-keeping.
2. **Observe:** track the bob's image position x(t) in a region of interest over 2-3 swings. Fit $x(t) = A \cdot \sin(2\pi t/T + \varphi 0)$ to obtain period, amplitude and phase with a confidence estimate.
3. **Predict:** compute the required clear interval t_req = t_delay + t_cross + t_margin, where t_cross is the time for the whole 0.70 m sphere to leave the hold line and clear the swing corridor. With the CAD-based launch acceleration of about 1.44 m/s^2 (pendulum at 60°), a 0.3 m hold distance, a 0.5 m corridor (assumed, Q6) and a 1.5 m/s cap, t_cross ≈ 1.5 s; with 0.15 s sensing delay and a 0.5 s margin, t_req ≈ 2.2 s. Find the next window in which the bob stays outside the corridor for t_req.
4. **Commit and cross:** start at the computed instant and do not stop or reverse inside the swing zone, because continuing clears it sooner than backing out. If confidence drops before the commit point, stay at the hold line and re-track. Holding is feasible because a level floor needs almost no drive torque to keep station. The crossing policy never overrides emergency stop or a detected safety fault; safety shutdown remains active in every operating state.

Pendulum geometry, period and position are not yet published (Q6). Algorithms are simulated in Phase 2 and validated with a test pendulum in Phase 3.

16. Emergency-stop and fail-safe concept

- **Transmitter:** handheld, battery-powered, latching mushroom switch, 2.4 GHz heartbeat at 20 Hz. It carries no motion commands.
- **Supervisor:** an independent MCU and receiver, powered from the logic rail. RUN = (valid heartbeat within 250 ms) AND (WD_OK) AND (not latched). Otherwise: **A** opens the normally-open motor-power contactor (all drivers and actuators lose power); **B** gates both drivers' PWM inputs to zero through a supervisor-controlled gate with pull-downs (the MDD20A and MD10C have PWM/DIR inputs only, no enable pin); **C** releases changeover brake relays: the normally-open side connects each drive motor to its driver, the normally-closed side shorts it through a resistor, and break-before-make means a driven output is never shorted. All three are fail-safe: loss of power or signal produces STOP.
- **Actuation versus motion:** STOP ends all motor actuation immediately. The sphere itself coasts, slowed by rolling resistance and the shorted motors, and can roll on a slope; Phase 2 quantifies the stopping distance.
- **Supervisor freeze:** the contactor and brake-relay coils are driven through pulse-driven (charge-pump) stages that stay energised only while the supervisor keeps toggling its outputs. A frozen supervisor, stuck high or low, lets every

coil drop out, which is STOP. Every safety output defaults to the de-energised (stopped) state at power-up and on any wiring break.

- **Status of this section:** a concept-level design. The detailed relay and gate schematic, and the measured stop latency, are Phase 2 deliverables.
- **Watchdog:** the motor MCU must toggle WDI. If it hangs, WD_OK drops and the supervisor stops the robot.
- **Latching and re-arm:** a stop persists until the transmitter is reset *and* re-arm is requested. The status LED in the left hub shows armed, e-stop or fault.
- **Manual disconnect:** the anti-spark arming plug in the left hub isolates the battery for handling, charging and inspection.
- **Latency budget (target, measured in Phase 2):** heartbeat timeout 250 ms; button-to-contactor under 100 ms when a STOP packet is received; contactor release about 10-20 ms (relay class).

Failure	Effect	Response
Transmitter battery dead or out of range	no heartbeat	STOP within 250 ms (fail-safe)
Pi 5 crash	no commands	MCU stops (200 ms time-out); e-stop still active
MCU crash	WDI stops	watchdog then supervisor: STOP
Supervisor brown-out	coil drive lost	contactor opens, brakes drop: STOP
Contactor welded	channel A lost	channels B + C still stop the motors; self-test at arming

17. Preliminary bill of materials and mass budget

Item	Description	Qty	Mass (kg)	PH
SHELL-HEMI	Clear polycarbonate hemisphere, OD 700 mm, 3 mm wall (R has 4 rim notches)	2	5.3968	no
SPIGOT-BAND	Equatorial spigot band + 4 key tabs, clear PC 5 mm	1	0.3903	no
PU-TAPE	Clear PU protective tape band, 0.35 mm (applied in narrow strips)	1	0.1848	no
POLE-FLANGE	Pole flange with bearing seat (PA12-CF print + Al insert)	2	0.4766	no
RING-GEAR	Ring gear 72T m1.5, face 12 mm (POM-C), bored	2	0.1880	no
POLE-BEARING	Deep-groove bearing 61908-2RS (6908), 40 x 62 x 12 mm	2	0.2200	no
AXLE	Stationary axle tube 30 x 3 mm, 6061-T6, 624 mm	1	0.4287	no
HUB-LEFT	Left polar hub (non-rotating, flush): cap, cup, boss	1	0.2196	no
HUB-RIGHT	Right polar hub (non-rotating, flush)	1	0.2236	no
ARMING-PLUG	Anti-spark arming plug + receptacle (XT90-S class) - manual disconnect	1	0.0300	yes
CHARGE-PORT	Charge port receptacle (BMS charge input)	1	0.0060	yes
STATUS-LED	Status LED (RGB, panel)	1	0.0020	yes
PRELOAD-BOLT	M12 preload bolt + disc-spring stack + axle end plug	1	0.0444	no
SIDE-PLATE	Stator side plate, 6061 3 mm, laser-cut	2	0.3988	no
END-PLATE	Stator front/rear plate, 6061 3 mm, laser-cut	2	0.3044	no
AXLE-CLAMP	Split axle clamp block, 6061	2	0.0966	no
GEARMOTOR	24 V brushed DC planetary gearmotor with Hall encoder, 42 mm class	2	1.2400	yes
PINION	Pinion 18T m1.5, steel	2	0.0676	no
MOTOR-ARM	Motor arm clamped to axle, 6061 6 mm (slotted)	2	0.1670	no
PIVOT-SHAFT	Steering pivot shaft, 12 mm steel	1	0.2566	no
PIVOT-BEARING	Pivot bearing 6001-2RS + housing	2	0.0700	no
STEER-MOTOR	Self-locking worm gearmotor (steering)	1	0.1600	yes
STEER-ANGLE	Magnetic angle sensor + magnet (steering pivot)	1	0.0040	yes
TILT-STOP	Rubber tilt end-stop	4	0.0064	no
CAMERA	Raspberry Pi Camera Module 3 Wide	1	0.0040	no
CAM-SERVO	Metal-gear micro servo (camera pitch)	1	0.0140	yes

Item	Description	Qty	Mass (kg)	PH
IMU	9-DoF IMU breakout	1	0.0060	yes
HALL-INDEX	Hall sensor + magnet (shell index)	1	0.0020	yes
PI5	Raspberry Pi 5, 8 GB + active cooler	1	0.0700	yes
MCU-BOARD	Motor-control MCU carrier board (RP2350 class)	1	0.0400	yes
DRIVE-DRIVER	Dual-channel brushed DC motor driver, 20 A/ch	1	0.0900	yes
STEER-DRIVER	Single-channel brushed DC motor driver, 13 A	1	0.0450	yes
PDB	Power distribution board: fuses, bus monitor, bus bars	1	0.0800	yes
DCDC-5V	DC-DC converter 24 V -> 5.1 V / 5 A	1	0.0350	yes
DCDC-6V	DC-DC converter 24 V -> 6 V / 3 A	1	0.0150	yes
CONTACTOR	Motor-power contactor / relay, NO, 24 V coil, 40 A	1	0.0400	yes
BRAKE-RELAYS	Dynamic-brake relays (NC) + braking resistors	1	0.0600	yes
ESTOP-SUPERVISOR	E-stop receiver + safety supervisor board + watchdog IC	1	0.0300	yes
HARNESS	Wiring harness and cable-routing allowance	5	0.3000	yes
FASTENERS	Fasteners, angle brackets, standoffs (allowance)	1	0.2500	yes
BOB-HANGER	Steering-bob hanger plate, 6061 5 mm	2	0.3192	no
BOB-TRAY	Battery tray, 6061 4 mm	1	0.1814	no
BATTERY	Battery 6S1P 21700 Li-ion + 6S BMS + wrap	1	0.5100	yes
BALLAST-PLATE	Ballast arc plate, laser-cut 8 mm steel, R215-R265 about the pivot axis	4	4.2088	no
BALLAST-BRACKET	Ballast clamp bracket + 2 x M6 tie-rods	2	0.1200	no
	Total on robot (equals CAD total)		**17.003**	
ESTOP-TX	Handheld wireless e-stop transmitter (latching mushroom switch, ESP32 class) (off-robot)	1	-	yes
CHARGER	6S Li-ion CC/CV charger, 25.2 V, <= 4.5 A (off-robot)	1	-	yes
SERVICE-STAND	Service stand (holds the ball left-hub-down for opening) (off-robot)	1	-	yes
GAUGES	Go/no-go ring gauges 2.0 ft and 2.5 ft (QC) (off-robot)	2	-	yes

The full list with CG coordinates, mass basis and placeholder flags is in **Preliminary_BOM.csv**. **No prices are given**: none have been quoted or verified yet. Placeholder items total 3.03 kg; their final masses are confirmed when parts are selected (Phase 2) and weighed (Phase 3).

18. Manufacturing and assembly approach

Part family	Process	Notes
PC hemispheres Ø700	purchased thermoformed domes	sourcing, wall uniformity and tolerance to be confirmed (risk R9)
Spigot band, key tabs	rolled and solvent-bonded PC strip	bonded into the left rim on a fixture
Stator plates, arms, hangers, tray	laser-cut 6061 / 5052 sheet, tapped	flat parts, DXF from CAD
Ballast arcs	laser-cut 8 mm mild steel	flat profiles; count tunes the CG
Axle, pivot shaft	cut tube / ground rod	flats for the keyed hubs
Pole flanges, hub caps	PA12-CF 3D print + aluminium inserts	bearing seats reamed after printing
Ring gears / pinions	stock m1.5 POM and steel gears, bored	concentricity to the bearing checked on the lathe

Assembly sequence: (1) Build the stator (plates, clamps, axle, pivot, bob, electronics) and test the electronics on the bench. (2) Fit the left flange and bearing to the left hemisphere; slide in the stator and fit the left hub. (3) Mesh the left gears. (4) Fit the right hemisphere; its ring gear meshes on assembly. (5) Torque the M12 preload bolt. (6) Apply the tape band. **Quality checks:** diameter with a go/no-go ring gauge (2.5 ft) and a 2.0 ft gauge; CG by a two-axis suspension test; static balance; e-stop functional test at every arming.

19. Main technical risks and mitigation strategies

ID	Risk	L	I	Mitigation (phase)
R1	Ramp steeper than the pendulum limit (10.7 deg baseline)	M	H	Ballast to 6 plates (13.6 deg incl. rolling resistance), run-up energy strategy (estimate only), tape grip; ramp geometry asked (Q3); Phase 2 dynamic model
R2	Entry lip or drop at the ramp ends stalls or shocks the robot	M	H	Approach speed, software torque limit, compliant gear mounts; question Q3; drop test in Phase 3
R3	Image quality through scratched/dirty PC, glare	M	M	Replaceable PU tape band, seam mask, adaptive thresholds + optional CNN, spare hemisphere; gyro/odometry bridging
R4	Pendulum pitch oscillation on accel/brake	H	M	Pitch-stabilised camera, active damping in the 500 Hz loop, speed ramps; Phase 2 model
R5	Lean steering too weak at low speed -> cone contacts	M	M	+/-22.9 deg static lean authority, early lateral corrections, speed planning; Phase 2 turning-radius model
R6	Barrier impact causes heading loss or damage	M	M	Heavy low CG, gyro heading hold, robust axle and bearings, torque limiting; Phase 2 impact loads
R7	Station-keeping before the pendulum (pendulum robots rock)	M	H	Encoder + IMU hold loop, hold line set back from the swing plane; Phase 2 validation
R8	E-stop link range or interference in the arena	L	H	Heartbeat fail-safe (loss = stop), internal antenna placed high behind the clear upper shell (no external mount), range test; sub-GHz fallback
R9	700 mm PC hemispheres: sourcing, wall uniformity, lead time	M	H	Early supplier enquiries; diameter is a parameter (re-run CAD); acrylic/PETG fallback
R10	Real masses/CG differ from estimates (3.0 kg placeholders)	M	M	Adjustable ballast, CG suspension test, parametric CAD update
R11	Pi 5 thermal throttling in a sealed shell	M	M	Active cooler + Al plate heat spreader, temperature monitoring; Phase 2 thermal estimate
R12	Gear mesh misalignment from flange compliance	L	M	Bearing-referenced flanges, slotted motor arms, POM ring gear; inspection at assembly
R13	Rule interpretation (diameter measurement, isolation, AI use)	M	H	Doubts-section questions Q1, Q5, Q8 before submission
R14	Li-ion pack safety under impact	L	H	Cylindrical cells in a rigid padded tray, BMS, 40 A fuse, fire-safe charging bag

20. Design assumptions and unresolved organizer questions

Assumptions: (A1) The track is essentially linear; turn radii are unknown. (A2) Track width of about 2 m is used only to illustrate the field of view. (A3) Ramp angles are unknown, so capability is shown from 5 to 35°. (A4) Rolling resistance c_rr = 0.03. (A5) Uniform 3 mm shell (real domes thin toward the pole). (A6) Purchased-part masses are estimates (PH, 3.03 kg). (A7) Gear stage efficiency 0.95. (A8) Indoor, hard and level floor with adequate lighting. (A9) A team-built wireless e-stop is acceptable. (A10) **No weight limit is stated, and none is assumed**; inspection records the weight.

Questions for the organizers (to be posted in the official Doubts section; full list in *Open_Questions_for_Organizers.md*): ramp angle, length, surface and entry/exit geometry; track and gate widths and cone size; pendulum size, period, amplitude and position; how the diameter is measured and whether limits are exclusive; the meaning of "electrical isolation" at inspection; kill-switch requirements; start procedure; whether AI-assisted documentation is acceptable and must be disclosed; and the Round 1 submission format and channel.

21. Future validation plan for Phase 2 (target 15 Nov 2026)

Phase 2 deliverable	Method	Acceptance target
Actuator torque + gear ratio for the 0.8 ft ramp	3-body dynamic model (shell, stator, bob) for ramps 5-35°, steps and stops; motor torque-speed curves	gearmotor selected with ≥ 1.5× margin at the worst case
Ramp feasibility and stability	simulation including run-up, entry lip and exit (down-slope or drop)	climbs all three ramps in simulation with the confirmed geometry
Balance stability	linearised pitch/roll model; LQR/PID damping design	pitch oscillation settles within 2 s
Gear, axle and bearing loads	hand calculation + FE for impact cases (0.8 ft drop, barrier hit)	SF ≥ 2 on yield, bearing life adequate
Battery and runtime	measured Pi 5 power + simulated motor currents	≥ 3 full runs plus 20 min reserve

Phase 2 deliverable	Method	Acceptance target
Kill switch	bench prototype; latency and fail-safe tests (TX off, interference, MCU hang)	stop < 300 ms in every case
Vision	cone and pendulum datasets; through-PC imaging test on a PC sample	lane estimate error < 0.15 m at 5 m
Pendulum crossing	Monte-Carlo timing simulation with assumed pendulum parameters	zero simulated collisions at the chosen margin
Full track	integrated simulation of the 75 m course	finishes without cone contact in simulation

Appendix A - Reproducing the CAD, checks and figures

```
cd cad
..\.venv\Scripts\python.exe build_all.py          # checks, STEP/STL/GLB, renders, drawings
cd ..\diagrams
..\.venv\Scripts\python.exe build_diagrams.py     # system / electronics / power / software diagrams
cd ..\docs_src
..\.venv\Scripts\python.exe build_docs.py         # regenerates both documents, CSVs and PDFs
```

Python 3.12 venv packages: cadquery 2.8, pyvista 0.49 / VTK 9.6, trimesh 5.1, manifold3d, shapely 2.2, matplotlib 3.11, markdown, python-docx, pymupdf. Diagrams and PDFs are rendered with headless Microsoft Edge.

Appendix B - Verification results (from cad/out/analysis_results.json)

Check	Result	Verdict
Diameter strictly within 609.6-762.0 mm (2562 directions)	696.3-700.7 mm	PASS
Same with +/-7 mm forming tolerance	689.3-707.7 mm	PASS
External features beyond the shell (limit 152.4 mm)	0 items; max -1.0 mm	PASS
Frame/bob to shell clearance, all pitch, bob +/-30 deg (>= 8 mm)	19.98 mm	PASS
Polar precision gaps (>= 1 mm)	2.0 mm (ST-02 vs SH-01)	PASS
B-rep clash check, 6 pitch/tilt states	0 clashes	PASS
Camera rays blocked by internal parts (< 0.5 %)	0.0 %	PASS
Battery removal path	clear	PASS
Preload-bolt socket access	clear	PASS
Lateral CG offset, abs(y) <= 2 mm	-0.4 mm	PASS

Appendix C - References

1. Techfest IIT Bombay, *The Wrecking Ball Challenge* problem statement (5 pp., 2026-27), and official page https://techfest.org/competitions/wrecking-ball-challenge with the Round 1 submission template (accessed 2026-10-20).
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15. IATA / CASA lithium-battery passenger guidance: ≤ 100 Wh without airline approval; 100-160 Wh with approval.